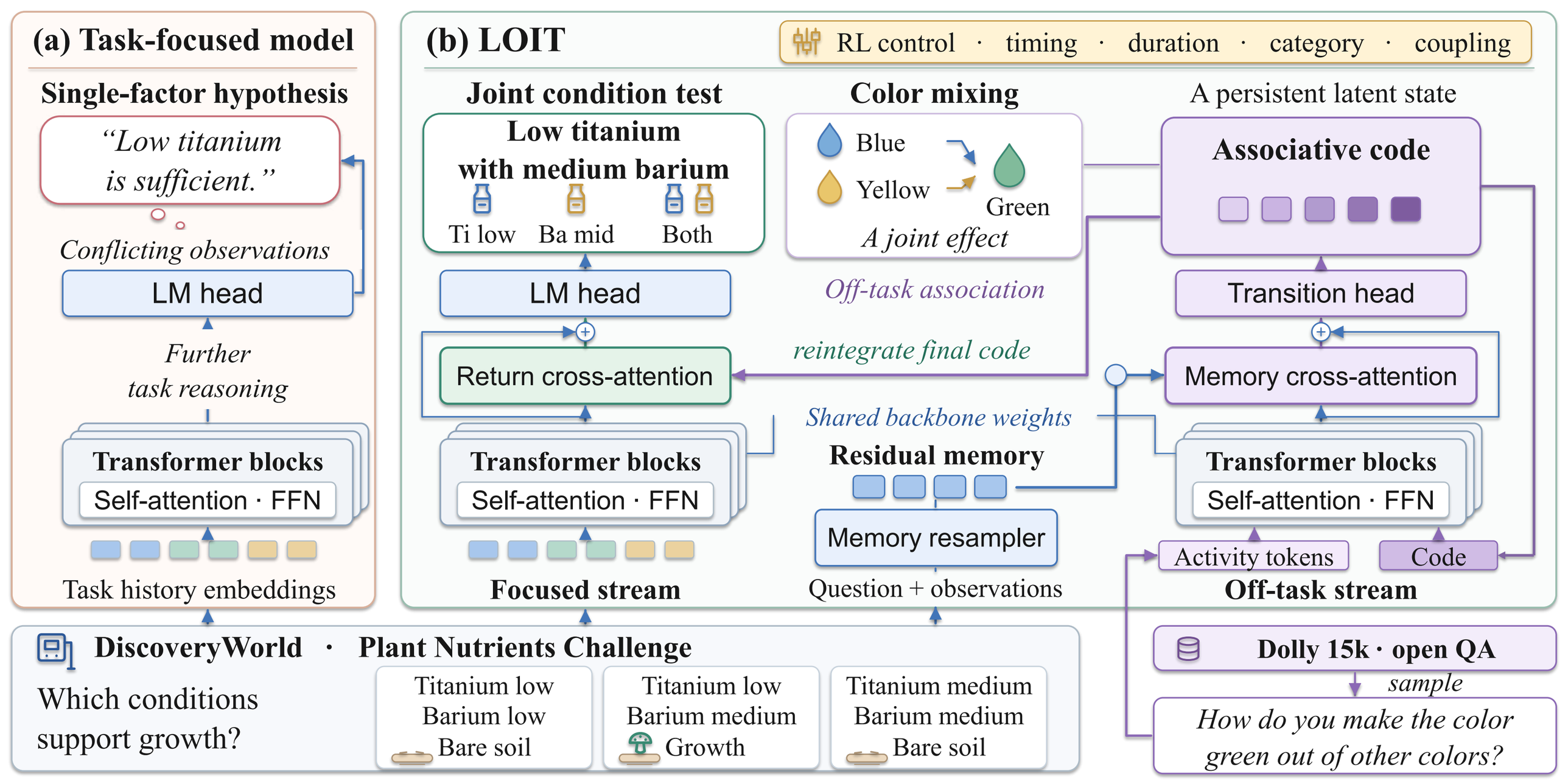

(a) Task-focused model
Single-factor hypothesis
"Low titanium is sufficient."
Conflicting observations
LM head
Further task reasoning
Transformer blocks
Self-attention · FFN
Task history embeddings
(b) LOIT
RL control · timing · duration · category · coupling
Joint condition test
Low titanium with medium barium
Ti low
Ba mid
Both
Color mixing
Blue
Yellow
Green
A joint effect
A persistent latent state
Associative code
LM head
Off-task association
Transition head
reintegrate final code
Return cross-attention
Memory cross-attention
Shared backbone weights
Transformer blocks
Self-attention · FFN
Residual memory
Transformer blocks
Self-attention · FFN
Memory resampler
Activity tokens
Code
Focused stream
Question + observations
Off-task stream
DiscoveryWorld · Plant Nutrients Challenge
Which conditions support growth?
Titanium low
Barium low
Bare soil
Titanium low
Barium medium
Growth
Titanium medium
Barium medium
Bare soil
Dolly 15k · open QA
sample
How do you make the color green out of other colors?